

AI RECRUITER ENGINE - SYSTEM ARCHITECTURE MAP

Just to clarify: this document represents a detailed high-level system map. Final scope, depth, and cost depend on prioritization and phase definition once we align on requirements.

OVERVIEW

This document maps the complete frontend and backend architecture for the AI Recruiter Engine, excluding AI/ML components which will be handled externally.

FRONTEND ARCHITECTURE

1. HR/Recruiter Portal

A. Vacancy Management Module

- **Create Vacancy Page**
 - Form: JD selection dropdown, role level, department
 - Validation: JD approval status check
 - Status management: Draft/Open/Closed toggle
- **Vacancy Dashboard**
 - List view: All vacancies with filters
 - Search & filter: By status, department, date
 - Quick actions: Edit, close, view candidates

B. Candidate Management Module

- **Candidate List View**
 - Ranked list display with scores
 - Filters: Experience level, score threshold, internal/external
 - Bulk actions: Export, send interview invitations
 - Sort options: By score, date, name
- **Candidate Profile View**
 - CV data display (structured)
 - Score breakdown visualization (charts/graphs)
 - Competency mapping table
 - Interview history timeline
 - Video interview recordings player
 - Interview analysis reports viewer

C. Interview Management Module

- **Interview Questions Generator**

- JD-based question generation trigger
- Questions preview/edit interface
- Difficulty level selector
- Export options (PDF/DOC download)
- **AI Avatar Interview Setup**
 - Interview invitation form
 - Email template customization
 - Schedule/deadline settings
 - Accessibility options configuration
- **Interview Analysis Dashboard**
 - Video playback with timestamps
 - Transcript viewer (synchronized with video)
 - Score visualization per competency
 - Communication metrics display
 - Sentiment indicators
 - Red flags highlighting
 - Comparative candidate view

D. Decision Support Module

- **Hiring Recommendation View**
 - Top candidates comparison table
 - AI reasoning display
 - Strengths/risks/gaps summary
 - Approve/override buttons
 - Override reason modal/form
 - Decision history log

E. Internal Mobility Module

- **Mobility Dashboard**
 - Open internal roles list
 - Employee matching view
 - Role-fit scores display
 - Skill gap analysis viewer
 - Recommendation reports

F. Succession Planning Module

- **Succession Matrix View**
 - Critical roles grid
 - Successor cards per role
 - Readiness level indicators (Ready/6-12mo/12+mo)
 - Performance data visualization
 - Edit succession plan interface

G. Promotion Module

- **Promotion Recommendations Dashboard**
 - Eligible employees list
 - Promotion scores display
 - Performance history viewer
 - Justification reports
 - Approval workflow

2. Candidate Portal

A. Interview Interface

- **Pre-Interview Page**
 - Identity validation form
 - Interview instructions
 - Technical setup check (camera/mic)
 - Privacy policy consent
 - **AI Avatar Interview Page**
 - Avatar video display (full screen option)
 - Question display area
 - Recording indicator
 - Pause/Resume button (once)
 - Progress bar
 - Timer display
 - Captions toggle
 - Response recording interface
 - **Post-Interview Page**
 - Thank you message
 - Next steps information
 - Contact support option
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BACKEND ARCHITECTURE

1. Core Services

A. API Gateway

- Request routing
- Authentication/Authorization
- Rate limiting
- Request logging
- CORS handling

B. Authentication Service

- User login/logout

- JWT token generation
- Role-based access control (HR, Recruiter, Candidate)
- Session management
- Password reset

2. Business Logic Services

A. Vacancy Service

- **Endpoints:**
 - POST /api/vacancies - Create vacancy
 - GET /api/vacancies - List vacancies
 - GET /api/vacancies/{id} - Get vacancy details
 - PUT /api/vacancies/{id} - Update vacancy
 - PATCH /api/vacancies/{id}/status - Change status
 - DELETE /api/vacancies/{id} - Delete vacancy
- **Business Logic:**
 - Validate JD approval status
 - Enforce mandatory fields
 - Status transition rules
 - Link to approved JD

B. Candidate Service

- **Endpoints:**
 - POST /api/candidates - Create candidate
 - POST /api/candidates/bulk-upload - Bulk CV upload
 - GET /api/candidates - List candidates
 - GET /api/candidates/{id} - Get candidate profile
 - GET /api/vacancies/{id}/candidates - Get candidates for vacancy
 - GET /api/candidates/{id}/ranked - Get ranked candidate list
- **Business Logic:**
 - Parse CV data (call AI service)
 - Store structured candidate data
 - Retrieve scoring from AI service
 - Ranking logic with tie-breakers
 - Filter and search candidates

C. Interview Service

- **Endpoints:**
 - POST /api/interviews/questions/generate - Generate questions
 - POST /api/interviews/avatar/invite - Send avatar interview invitation
 - GET /api/interviews/{id} - Get interview details
 - POST /api/interviews/{id}/start - Start interview session

- POST /api/interviews/{id}/responses - Submit response
- PATCH /api/interviews/{id}/pause - Pause interview
- PATCH /api/interviews/{id}/resume - Resume interview
- POST /api/interviews/{id}/complete - Complete interview
- GET /api/interviews/{id}/analysis - Get interview analysis
- GET /api/interviews/{id}/transcript - Get transcript
- GET /api/interviews/{id}/video - Stream video recording
- **Business Logic:**
 - Generate role-specific questions (call AI service)
 - Send email invitations
 - Validate candidate identity
 - Manage interview state (started/paused/completed)
 - Store video/audio recordings
 - Request AI analysis
 - Generate analysis reports
 - Secure video access control

D. Decision Service

- **Endpoints:**
 - GET /api/decisions/recommendations/{vacancyId} - Get hiring recommendations
 - POST /api/decisions/{id}/approve - Approve recommendation
 - POST /api/decisions/{id}/override - Override with reason
 - GET /api/decisions/{id}/history - Get decision history
- **Business Logic:**
 - Combine CV + Interview scores
 - Generate recommendation summary
 - Capture override reasons
 - Log decision history

E. Internal Mobility Service

- **Endpoints:**
 - GET /api/mobility/roles - List internal roles
 - POST /api/mobility/scan - Scan employee profiles
 - GET /api/mobility/{roleId}/matches - Get matched employees
 - GET /api/mobility/reports/{employeeId} - Get mobility report
- **Business Logic:**
 - Match employee profiles to roles
 - Calculate role-fit scores (call AI service)
 - Identify skill gaps
 - Generate recommendation reports

F. Succession Planning Service

- **Endpoints:**

- POST /api/succession/roles - Define critical role
- GET /api/succession/roles - List critical roles
- GET /api/succession/{roleId}/successors - Get successors
- POST /api/succession/{roleId}/identify - Identify successors
- GET /api/succession/matrix - Get succession matrix
- **Business Logic:**
 - Identify successors (call AI service)
 - Assign readiness levels
 - Generate succession matrix
 - Link to performance data

G. Promotion Service

- **Endpoints:**
 - POST /api/promotions/initiate - Start promotion cycle
 - GET /api/promotions/eligible - Get eligible employees
 - POST /api/promotions/{employeeId}/evaluate - Evaluate employee
 - GET /api/promotions/{employeeId}/recommendation - Get recommendation
 - POST /api/promotions/{id}/approve - Approve promotion
- **Business Logic:**
 - Validate eligibility rules
 - Retrieve performance history
 - Calculate promotion scores (call AI service)
 - Generate justification reports

3. Integration Services

A. Job Description Integration Service

- **Endpoints:**
 - GET /api/jd/{id} - Fetch JD details
 - GET /api/jd/{id}/competencies - Get competencies
 - GET /api/jd/approved - List approved JDs
- **Business Logic:**
 - Integrate with JD Engine
 - Validate JD approval status
 - Extract competencies

B. ATS Integration Service (if applicable)

- **Endpoints:**
 - POST /api/ats/sync - Sync candidates from ATS
 - GET /api/ats/candidates - Fetch ATS candidates
- **Business Logic:**
 - Connect to external ATS

- Map ATS data to internal format
- Sync candidate updates

C. Email Service

- **Endpoints:**
 - POST `/api/notifications/email` - Send email
 - POST `/api/notifications/template` - Send templated email
- **Business Logic:**
 - Send interview invitations
 - Send reminders
 - Send result notifications
 - Template management

D. Document Service

- **Endpoints:**
 - POST `/api/documents/upload` - Upload CV/document
 - GET `/api/documents/{id}` - Download document
 - POST `/api/documents/export` - Export reports (PDF/DOC)
- **Business Logic:**
 - File upload handling
 - File type validation
 - Secure storage
 - Document generation (PDF/DOC)

E. Video/Media Service

- **Endpoints:**
 - POST `/api/media/upload` - Upload video/audio
 - GET `/api/media/{id}/stream` - Stream video
 - POST `/api/media/{id}/transcribe` - Request transcription
- **Business Logic:**
 - Video/audio upload
 - Secure cloud storage
 - Streaming with authentication
 - Transcription request handling
 - Encryption for sensitive data

4. Data Services

A. Employee Data Service

- **Endpoints:**
 - GET `/api/employees` - List employees
 - GET `/api/employees/{id}` - Get employee profile
 - GET `/api/employees/{id}/performance` - Get performance data

- GET /api/employees/{id}/competencies - Get competencies
- **Business Logic:**
 - Retrieve employee profiles
 - Fetch performance data
 - Access competency framework

B. Competency Framework Service

- **Endpoints:**
 - GET /api/competencies - List all competencies
 - GET /api/competencies/role/{roleId} - Get role competencies
 - GET /api/competencies/{id} - Get competency details
- **Business Logic:**
 - Manage competency library
 - Map competencies to roles

5. AI Service Interface Layer

A. AI Proxy Service

- **Purpose:** Interface between backend and external AI services
- **Endpoints:**
 - POST /api/ai/cv-parse - Parse CV
 - POST /api/ai/score-candidate - Score candidate
 - POST /api/ai/generate-questions - Generate interview questions
 - POST /api/ai/analyze-interview - Analyze interview responses
 - POST /api/ai/match-mobility - Match internal mobility
 - POST /api/ai/identify-successors - Identify successors
 - POST /api/ai/calculate-promotion - Calculate promotion score
- **Business Logic:**
 - API calls to external AI services
 - Request/response mapping
 - Error handling for AI services
 - Caching AI results
 - Rate limiting for AI calls

DATABASE ARCHITECTURE

1. Core Tables

A. Vacancies

- id (PK)
- jd_id (FK to JD Engine)
- title
- role_level

- department
- status (Draft/**Open**/Closed)
- created_by
- created_at
- updated_at

B. Candidates

- **id** (PK)
- name
- email
- phone
- **source** (External/Internal)
- cv_file_path
- parsed_data (JSON)
- created_at
- updated_at

C. Candidate__Vacancy (Junction Table)

- **id** (PK)
- candidate_id (FK)
- vacancy_id (FK)
- cv_score
- ai_reasoning (JSON)
- rank_position
- status (**New**/Shortlisted/Interviewed/Rejected/Hired)
- applied_at

D. Interviews

- **id** (PK)
- candidate_id (FK)
- vacancy_id (FK)
- **type** (Traditional/AI_Avatar)
- status (Scheduled/In_Progress/Paused/Completed)
- scheduled_at
- started_at
- completed_at
- video_file_path
- transcript_file_path
- analysis_data (JSON)

E. Interview__Questions

- **id** (PK)
- interview_id (FK)

- vacancy_id (FK)
- question_text
- competency_mapped
- difficulty_level
- order_sequence
- generated_at

F. Interview_Responses

- id (PK)
- interview_id (FK)
- question_id (FK)
- response_text (transcript)
- video_timestamp_start
- video_timestamp_end
- score
- ai_feedback (JSON)
- created_at

G. Hiring_Decisions

- id (PK)
- vacancy_id (FK)
- candidate_id (FK)
- recommendation_type (Hire/Reject/Hold)
- ai_reasoning (JSON)
- hr_decision (Approved/Overridden)
- override_reason
- decided_by
- decided_at

H. Internal_Mobility

- id (PK)
- employee_id (FK)
- vacancy_id (FK)
- role_fit_score
- skill_gaps (JSON)
- recommendation
- created_at

I. Succession_Plans

- id (PK)
- critical_role_id
- successor_employee_id (FK)
- readiness_level (Ready/6-12mo/12+mo)

- competency_data (JSON)
- created_at
- updated_at

J. Promotions

- id (PK)
- employee_id (FK)
- current_role
- proposed_role
- promotion_score
- eligibility_status
- justification (JSON)
- status (Pending/Approved/Rejected)
- decided_at

2. Supporting Tables

K. Users

- id (PK)
- username
- email
- password_hash
- role (HR/Recruiter/Candidate)
- is_active
- created_at

L. Audit_Logs

- id (PK)
- user_id (FK)
- action
- entity_type
- entity_id
- changes (JSON)
- timestamp

M. Notifications

- id (PK)
- recipient_id (FK)
- type (Email/SMS/In_App)
- subject
- content
- status (Sent/Failed/Pending)
- sent_at

EXTERNAL INTEGRATIONS

1. JD Engine

- Purpose: Fetch approved job descriptions
- Integration Type: REST API
- Authentication: API Key / OAuth

2. AI/ML Services (External - Not Implemented Here)

- CV Parsing
- Candidate Scoring
- Interview Question Generation
- Interview Analysis
- Mobility Matching
- Succession Identification
- Promotion Calculation

3. Cloud Storage (Video/Documents)

- Provider: AWS S3 / Azure Blob / Google Cloud Storage
- Purpose: Store videos, CVs, reports
- Security: Encryption at rest and in transit

4. Email Service

- Provider: SendGrid / AWS SES / Mailgun
- Purpose: Interview invitations, notifications

5. Calendar Integration (Optional)

- Provider: Google Calendar / Outlook
 - Purpose: Schedule interviews
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SECURITY REQUIREMENTS

Backend Security

- JWT-based authentication
- Role-based access control (RBAC)
- API rate limiting
- Input validation and sanitization
- SQL injection prevention
- XSS protection

- HTTPS enforcement
- Data encryption at rest and in transit
- Secure video/document storage
- GDPR compliance for personal data
- Audit logging for all critical actions

Frontend Security

- Secure token storage
 - CSRF protection
 - Content Security Policy (CSP)
 - Secure video streaming
 - No sensitive data in client-side storage
 - Input validation before API calls
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NON-FUNCTIONAL REQUIREMENTS

Performance

- API response time: < 500ms (95th percentile)
- Video streaming: Adaptive bitrate
- Bulk CV upload: Support up to 100 CVs
- Concurrent interviews: Support 50+ simultaneous sessions

Scalability

- Horizontal scaling for API services
- Database read replicas
- CDN for video streaming
- Queue-based processing for heavy tasks (CV parsing, AI scoring)

Availability

- 99.9% uptime SLA
 - Database backups every 24 hours
 - Disaster recovery plan
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